



Threat Prevention

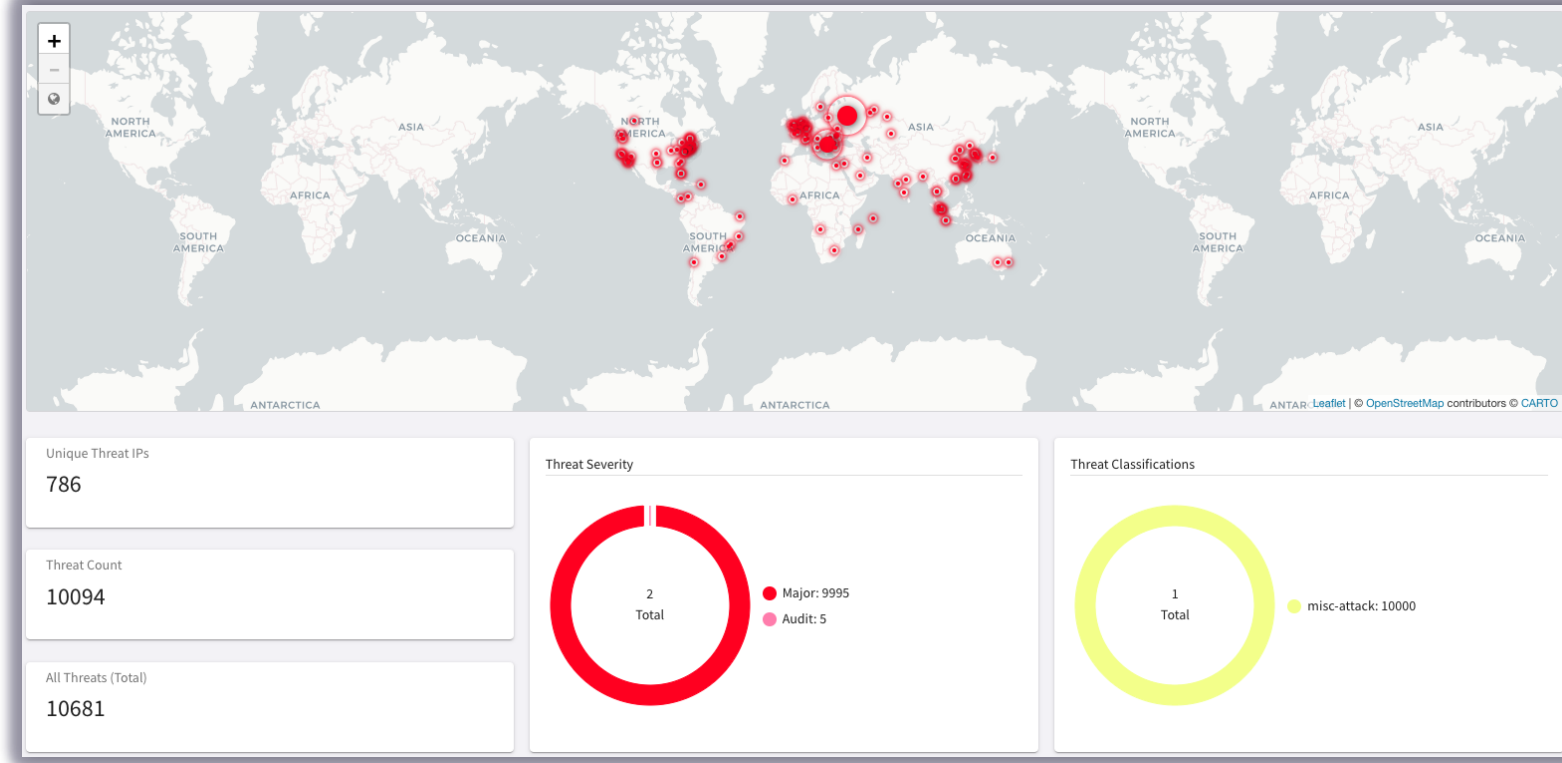
IDENTIFY AND REMEDIATE THREATS ACROSS MULTICLOUD NETWORKS

ACE Team

What is it?



- Multicloud native network security to dynamically **identify, alert, and remediate potential threats** to known malicious IP addresses
- **Distributed threat visibility** and control built into the Distributed Cloud Firewall service using the *ThreatGroup*
- Identify potential **data exfiltration and compromised host**
- **Complementary security solution** with full multicloud support



How does it work?

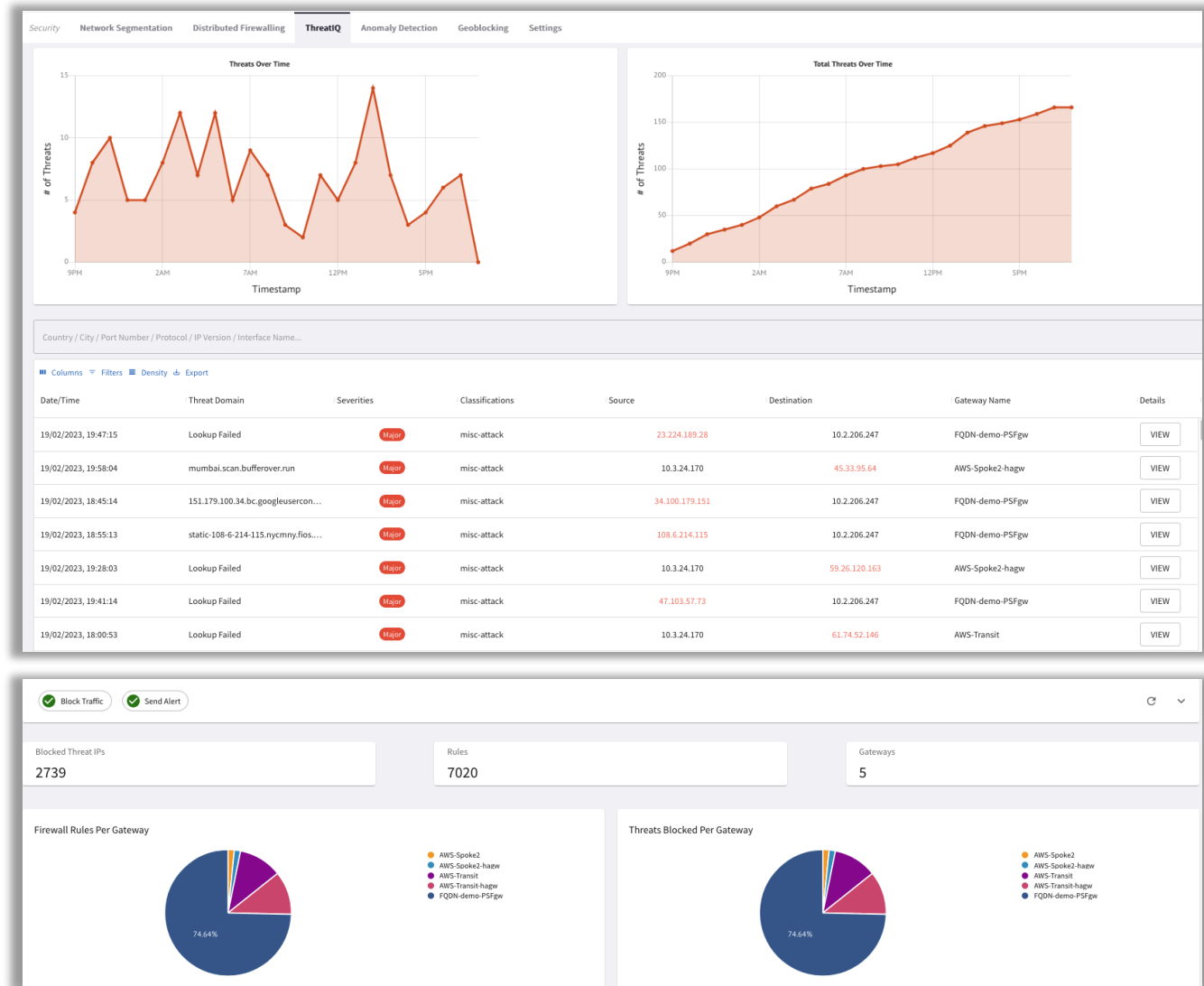


● Distributed Inspection & Notification

- Aviatrix gateways across Multicloud environment send real-time NetFlow data to CoPilot
- CoPilot analyzes the data on all public destinations against well-known Threat DB.
- CoPilot alerts on any potential threats in the environment
- CoPilot provides extreme visibility of the impacted communication flow

● Distributed Enforcement

- CoPilot informs Aviatrix Controller to push firewall policies to all the Aviatrix gateways in the data path
- Firewall policies automatically get updated with the current status of the threat.
- Blocking threats with firewall policy is optional but recommended



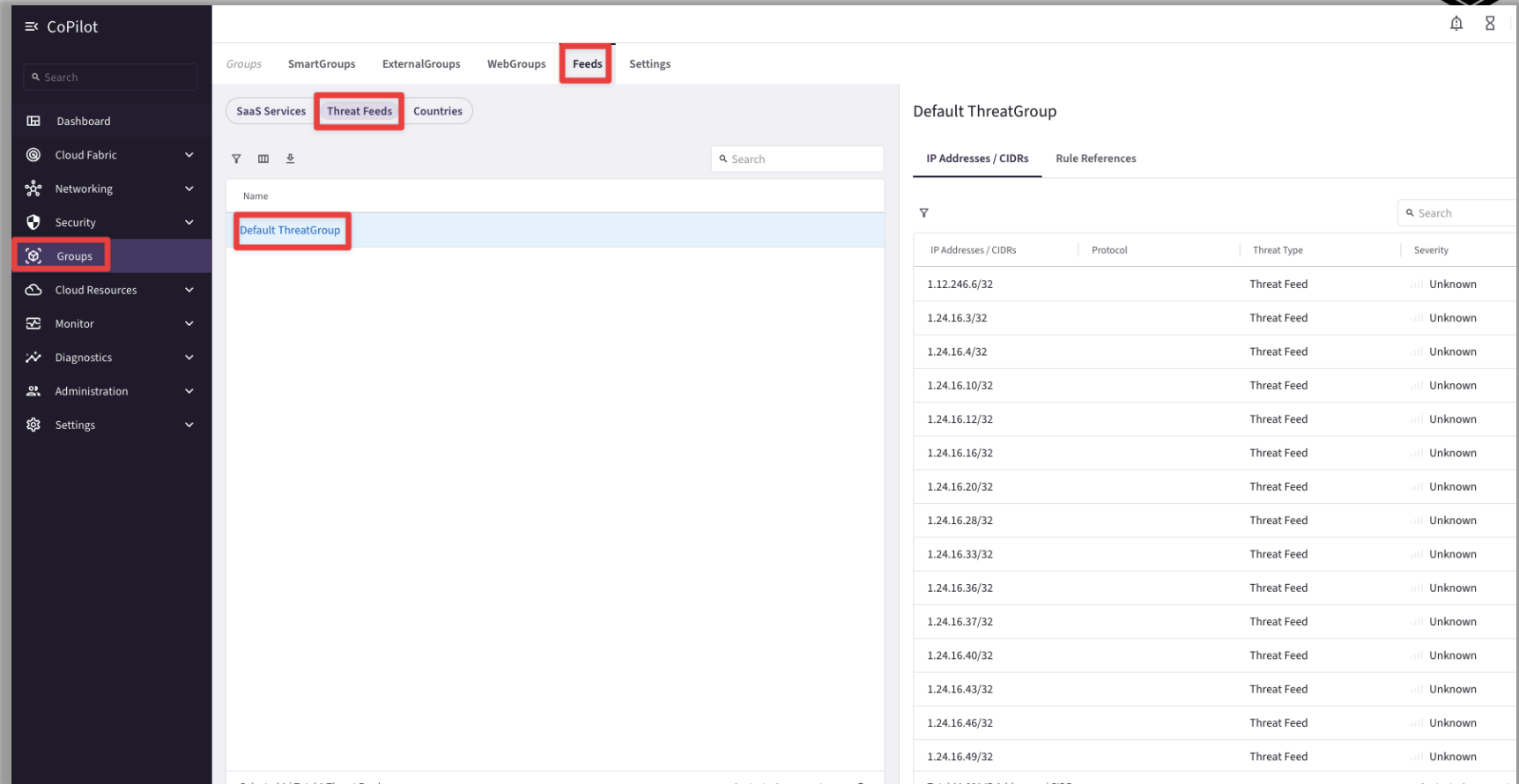
Default ThreatGroup

ProofPoint Database

- The **Default ThreatGroup** can be used to ensure that traffic meeting the ThreatGroup criteria is blocked
- The **Default ThreatGroup** is regularly updated with data from *ProofPoint Global Threat Defense Database* (every 30 min)
- The Default ThreatGroup references the complete list of all the Malicious IP addresses.

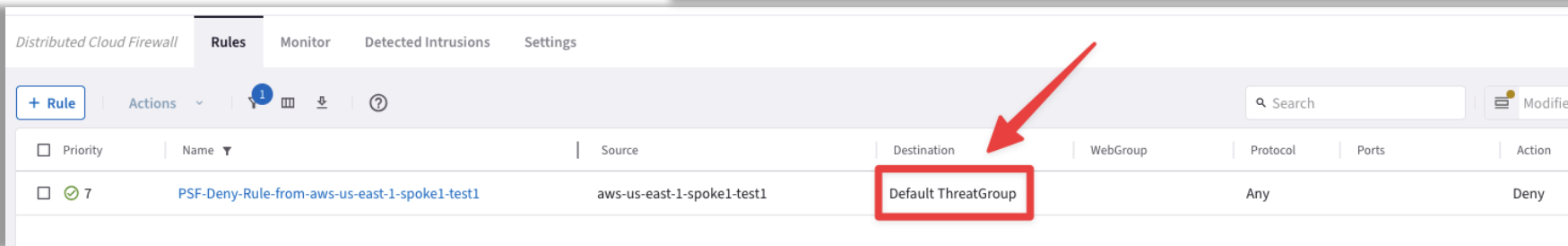
Note:

- You cannot have a ThreatGroup as both source and a destination in a DCF rule



The screenshot shows the Aviatrix CoPilot interface. On the left is a sidebar with navigation options: Dashboard, Cloud Fabric, Networking, Security, Groups (highlighted), Cloud Resources, Monitor, Diagnostics, Administration, and Settings. The main panel has tabs for Groups, SmartGroups, ExternalGroups, WebGroups, Feeds (highlighted), and Settings. Under the Feeds tab, there are sub-tabs for SaaS Services, Threat Feeds (highlighted), and Countries. A table lists various threat feeds, with 'Default ThreatGroup' highlighted. On the right, a detailed view of the 'Default ThreatGroup' is shown, displaying a table of IP addresses and their associated threat types and severities.

IP Addresses / CIDRs	Protocol	Threat Type	Severity
1.12.246.6/32		Threat Feed	Unknown
1.24.16.3/32		Threat Feed	Unknown
1.24.16.4/32		Threat Feed	Unknown
1.24.16.10/32		Threat Feed	Unknown
1.24.16.12/32		Threat Feed	Unknown
1.24.16.16/32		Threat Feed	Unknown
1.24.16.20/32		Threat Feed	Unknown
1.24.16.28/32		Threat Feed	Unknown
1.24.16.33/32		Threat Feed	Unknown
1.24.16.36/32		Threat Feed	Unknown
1.24.16.37/32		Threat Feed	Unknown
1.24.16.40/32		Threat Feed	Unknown
1.24.16.43/32		Threat Feed	Unknown
1.24.16.46/32		Threat Feed	Unknown
1.24.16.49/32		Threat Feed	Unknown



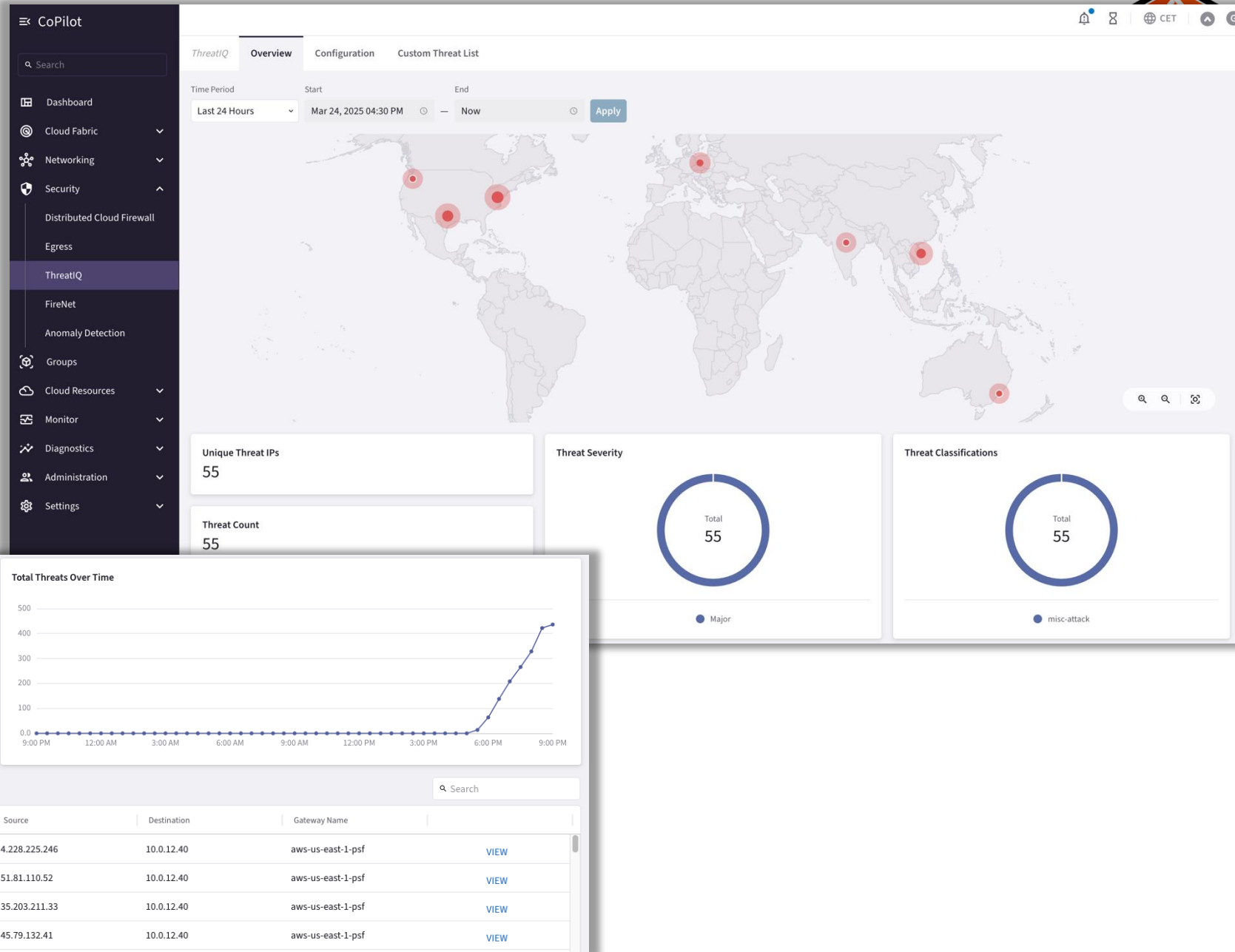
The screenshot shows the 'Rules' configuration page in the Distributed Cloud Firewall. A table lists firewall rules. The first rule, 'PSF-Deny-Rule-from-aws-us-east-1-spoke1-test1', has its 'Destination' field set to 'Default ThreatGroup', which is highlighted with a red box and a red arrow pointing to it.

Priority	Name	Source	Destination	WebGroup	Protocol	Ports	Action
7	PSF-Deny-Rule-from-aws-us-east-1-spoke1-test1	aws-us-east-1-spoke1-test1	Default ThreatGroup		Any		Deny

ThreatIQ

Overview Tab

- Shows a geographical map with the approximate locations of known malicious IPs that have communicated with your network within the specified time period selected.
- You can view the severity level of detected threat IPs and their associated attack classifications (as categorized by the well-known threat IPs DB).





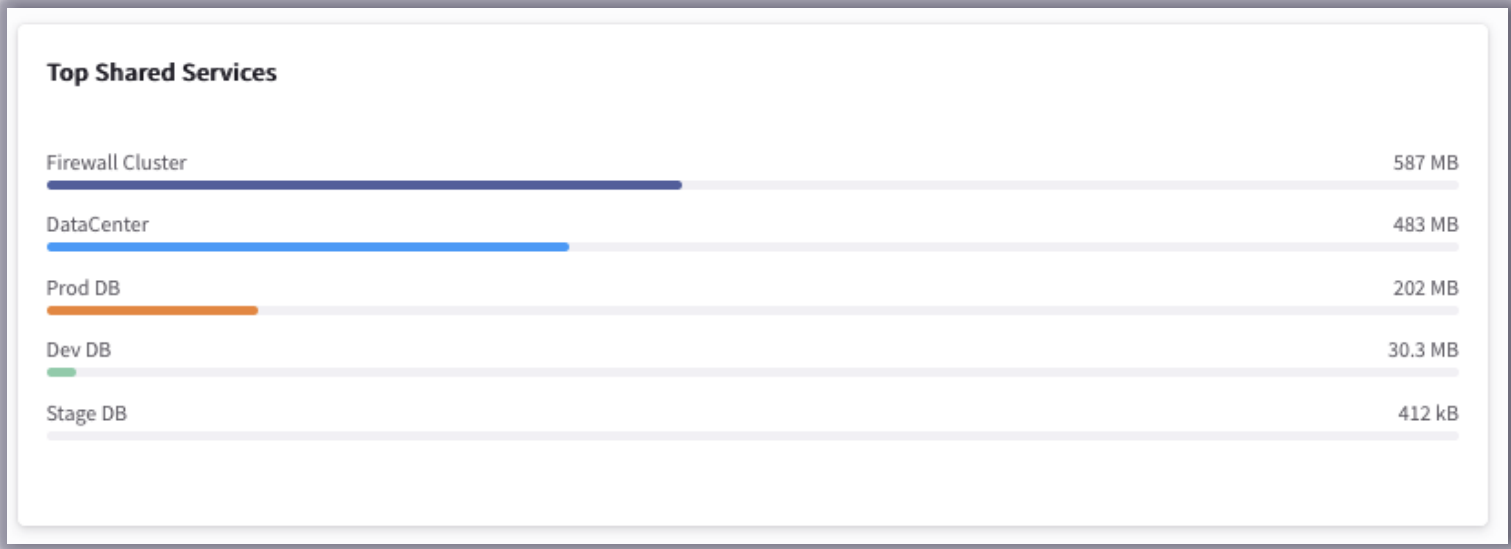
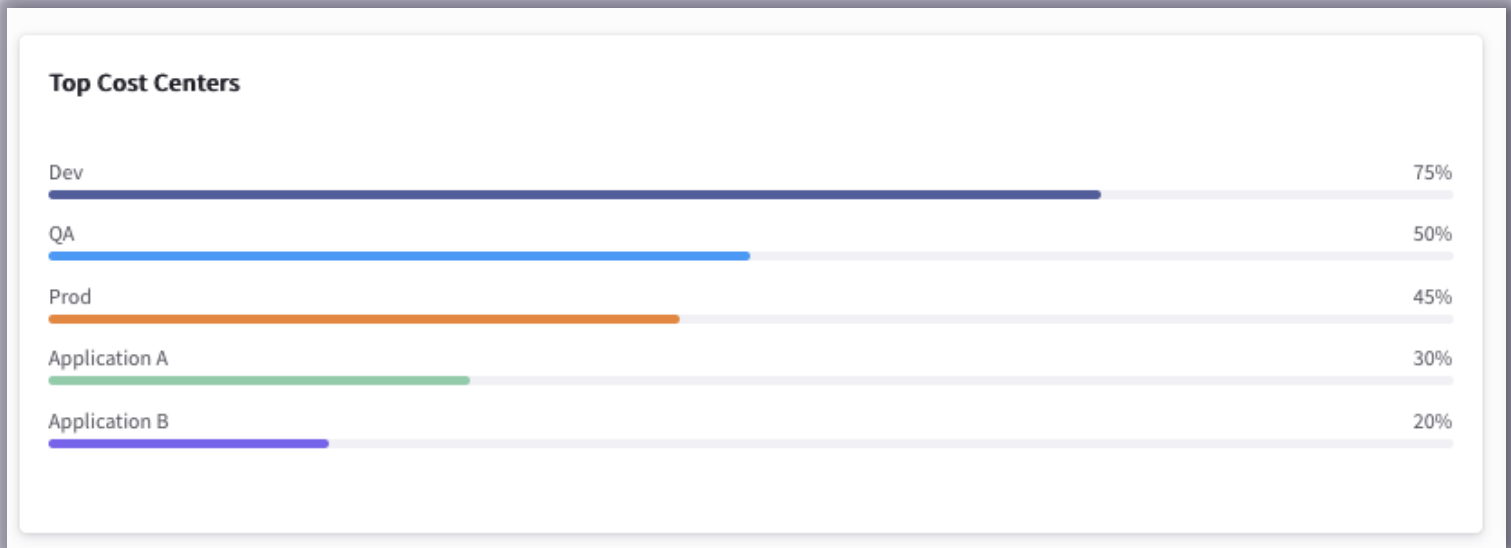
CostIQ

MONITORING THE COST OF YOUR BUSINESS UNITS

ACE Team

What is it?

- The **CostIQ** feature provides detailed traffic distribution analysis for your cost centers, including traffic flowing to shared-service resource hosts by Cloud Account, by Cost Center, by VPC/VNet, and by Gateway.
- The cost information displayed in CostIQ is grouped by:
 - **Cost Center** - A group of resources categorized by CSP (Cloud Service Provider) tags, associated VPCs/VNets. These CoPilot Cost Centers contain resources used by your real-life cost centers or business units.
 - **Shared Service** - A cloud or network resource shared by multiple teams or cost centers. You define Shared Services by listing the IP addresses or IP CIDR ranges of the shared resource hosts.



Cost Center (part.1)

CostIQ Overview **Cost Centers** Shared Services

The page below is a demo and shown with sample data.

+ Cost Center [Filter] [Grid] [Download] [Search]

Name	Clouds	VPC/VNets	Last 7 Days	Prev Week	Prev Month	Prev Quarter	MTD	QTD
Dev	GCP, Azure ARM	2	75%	75%	75%	75%	75%	75%
QA	AWS	1	50%	50%	50%	50%	50%	50%
Prod	Alibaba Cloud, Azure ARM	2	45%	45%	45%	45%	45%	45%
Application A	GCP, Azure ARM	2	30%	30%	30%	30%	30%	30%
Application B	GCP, Azure ARM, Alibaba Cloud	3	20%	20%	20%	20%	20%	20%

- The **Cost Center** is a logical grouping that represents a Line of Business or a department. Essentially, the Cost Center can embrace multiple VPCs/VNets across multiple clouds and multiple accounts.

Create Cost Center

Name

TEST

Associate VPC/VNets

aws-us-east-1-spoke1 × aws-us-east-2-spoke1 ×

Cancel Save

Cost Center (part.2)



CostIQ

Overview

Cost Centers

Shared Services

The page below is a demo and shown with sample data.

Enable CostIQ

+ Cost Center

Search

Name	Clouds	VPC/VNets	Last 7 Days	Prev Week	Prev Month	Prev Quarter	MTD
Dev	GCP, Azure ARM	2	75%	75%	75%	75%	
QA	AWS	1	50%	50%	50%	50%	
Prod	Alibaba ... , + 1 more	2	45%	45%	45%	45%	
Applicati...	GCP, Azure ARM	2	30%	30%	30%	30%	
Applicati...	GCP, + 2 more	3	20%	20%	20%	20%	

Prod

Time Period

Last 7 Days

Start

May 13, 2024 12:00 AM

End

May 20, 2024 12:00 AM

All Traffic *

Total

57.7 MB

Search

VPC/VNET	Region	Rel. Traffic	Traffic
aws-us-east-spoke1	us-east-1	39%	35.3 MB
aws-us-west-spoke1	us-west-2	61%	22.3 MB

- After defined a Cost Center, you can investigate all the associated Application VPCs/VNets that are all part of that Cost Center. You can drill down and find out the **relative amount of traffic** for each Application VPC/Vnet.

Shared Center (part.1)

+ Shared Service Filter View Download Search							
Name	IP or CIDRs	Last 7 Days	Prev Week	Prev Month	Prev Quarter	MTD	QTD
Firewall Cluster	10.11.1.0	587 MB	587 MB	587 MB	587 MB	587 MB	587 MB
Data Center	11.100.0.0/24	483 MB	483 MB	483 MB	483 MB	483 MB	483 MB
Prod DB	120.20.0.24	202 MB	202 MB	202 MB	202 MB	202 MB	202 MB
Dev DB	10.21.1.89, 10.21.1.50, 10.21.1.10	30.3 MB	30.3 MB	30.3 MB	30.3 MB	30.3 MB	30.3 MB
Stage DB	10.21.1.90	412 kB	412 kB				

Add Shared Service

Name

IP CIDRs

Cancel
Save

- The **Shared Service** is another logical grouping that represents a Shared Application, for instance a syslog collector like Splunk. You can also associate S3 buckets to your Shared Services.
- The Shared Service allows you to monitor the resources that try reaching your shared applications

Shared Center (part.2)

Cost/Q

Overview

Cost Centers

Shared Services

+ Shared Service

Filter

Grid

Download

Search

Name	IP or CIDRs	Last 7 Days	Prev Week	Prev Month	Prev Quarter	MT
Shared S3 Bucket	10.4.4.0/24	93.3 MB	92.9 MB	319 MB	620 MB	
Direct Connect Seattle	10.4.3.10/24	96.5 MB	95.7 MB	300 MB	664 MB	
Direct Connect Ashburn	10.3.2.0/24	2.74 GB	2.59 GB	17.1 GB	41.4 GB	
Shared Storage	10.4.2.0/24	92.6 MB	92 MB	321 MB	614 MB	

Shared S3 Bucket

Time Period

Last 7 Days

Start

May 13, 2024 12:00 AM

End

May 20, 2024 12:00 AM

Amount for Time Period

Enter cost

\$

Breakdown by

Co...

All Traffic *

Total

93.3 MB

Filter

Grid

Download

Search

Cost Center	Rel. Traffic	Traffic
Enterprise Data	97%	91.2 MB
Accounting	1.1%	1.07 MB
Marketing	1.1%	1.04 MB
Engineering	0%	0 B
Operations	0%	0 B
Total 5 Cost Centers		

- After defining a **Shared Service**, you can accurately find out what LOB/Department has been utilizing it.



Next: Lab 9 – Threat Prevention &
Lab 10 - CostIQ